# A Novel Metamodel-based Equilibration Scheme for Pricing Problems with Multiple Decision Makers in Full-Scale Traffic Models.

**Gaurav Malik**
FWO PhD SB Research Fellow,
Department of Mechanical Engineering
KU Leuven, Leuven, Belgium, 3001
Email: gaurav.malik@kuleuven.be
ORCiD: https://orcid.org/0000-0002-1978-0444

**Chris M.J. Tampère**
Professor,
Department of Mechanical Engineering
KU Leuven, Leuven, Belgium, 3001
Email: chris.tampere@kuleuven.be
ORCiD: https://orcid.org/0000-0002-5489-7816

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*Gaurav Malik

## ABSTRACT

Designing efficient road pricing schemes with multiple decision makers is computationally challenging, especially when computing game-theoretical equilibria using detailed traffic models. This paper presents a novel metamodel-based equilibration (MBE) scheme that leverages a fast game-theoretical metamodel, based on the CHANAkYA framework, to compute optimal or equilibrium tolls for a full-scale traffic model.

Unlike typical metamodel-based optimization methods, which approximate only the objective function of an individual decision maker, the MBE metamodel captures system-wide strategic interactions, enabling the computation of tolling equilibria across multiple decision makers. The scheme operates via an iterative calibration–equilibration cycle: the metamodel is calibrated using outputs from the detailed traffic model, followed by game-theoretical equilibration at the metamodel level, with updated tolls fed back into the detailed model.

We apply the method to single- and multi-player pricing scenarios on a real-world network with variable demand. The MBE scheme outperforms benchmark Bayesian Optimization in terms of both quality of tolls and computational efficiency and, for the first time, enables the computation of approximate Nash and Stackelberg equilibria in a full-scale traffic model with variable demand. These findings confirm that properly calibrated transport economics and mathematical programming-based models can produce equilibrium tolls that are effective in detailed traffic models.


**Keywords:** Road pricing optimization; transport economics; toll competition; metamodel-based optimization.

## INTRODUCTION

The transportation ecosystem includes multiple stakeholders whose decisions influence each other. For credible scenario prediction and informed policymaking, especially in road pricing, it is essential to model all relevant stakeholders. Road pricing decisions are rarely unilateral; they involve strategic interactions among governments at the same level (e.g., national or municipal) or different levels (e.g., regional and federal). For instance, recent tensions between the U.S. federal government, the State of New Jersey, and the State of New York over New York's congestion toll highlight the complexity of such policies.

Three primary branches of transportation research address congestion and pricing but fall short in resolving multi-decision-maker pricing problems in full-scale traffic models.

The first branch relies on **detailed traffic models**, including Static or Dynamic Traffic Assignment (STA/DTA) and Multi-Agent Transport Simulation (MATSim) [1]. These models offer high-fidelity traffic representations but are computationally prohibitive for multi-decision-maker pricing optimization. For example, MATSim has been used in [2] to assess cordon toll impacts in Brussels, but such studies used fixed toll values at predetermined locations. Even optimizing a single toll instrument in such settings is challenging. Identifying game-theoretical equilibrium involving multiple decision makers, each optimizing their own tolling strategies, is nearly infeasible.

A second branch emerged from the development of the Macroscopic Fundamental Diagram (MFD), including both the accumulation-based MFDs [3] and the trip-based MFD [4]. By assuming a pre-estimated MFD to represent aggregate regional congestion, these models simplify congestion representation and allow for faster toll optimization. For example, [5] integrated an MFD-based congestion model with a logit model-based departure-time choice model and employed Bayesian Optimization (BO) to optimize toll profiles. However, MFD-based models face structural limitations. Spatial resolution is lost due to aggregation, making it difficult to model route choice or spatially differentiated tolls which can be important in heterogeneous urban areas. Additionally, the MFD curve must be pre-estimated and may become outdated with infrastructural changes in the network. [6] highlights such limitations and falls back on a detailed traffic simulation (mesoscopic) of the greater Melbourne in AIMSUN. They used the surrogate-based optimization (regressing Kriging) for optimizing ten toll variables. However, this required 90 evaluations of the expensive underlying simulation for a single decision maker setup. Extending such approaches to find game-theoretical equilibria among multiple decision makers remains computationally infeasible.

The third branch stems from **transport economics and mathematical programming**. These models are generally more suitable for analyzing scenarios with multiple decision makers. Game-theoretical interactions between governments, both at the same level and at different levels, and in serial as well as parallel network configurations, have been studied in [7] and [8]. Their strength lies in reducing spatial detail to preserve the key strategic interactions among actors. However, these models often face criticism for their overly simplistic congestion and route-choice equilibrium modeling, limited user heterogeneity, and lack of applicability to real-world settings. Recent works like [9], [10], and [11] address these criticisms by incorporating mathematical programming techniques. [10] propose an optimization-based game-theoretical model of road pricing, which incorporates variable user demand, and an endogenized user-equilibrium, whereas [11] extend that model to a comprehensive framework (CHANAkYA) for pricing and tradable credit charging in multi-modal transportation systems including heterogeneous travelers, government authorities and private mobility service providers. Yet, such models typically rely on pseudo-real networks and simplified link cost functions, sometimes even linear, which raises concerns about their applicability to real-world contexts.

Despite progress in each branch, no approach has successfully computed game-theoretical equilibria in multi-decision-maker tolling problems using full-scale traffic models. Nevertheless, advances continue. [12] and [13] advance the development of multi-region MFDs incorporating a regional path-choice. [14], on the other hand, focus on improving BO by using contextual information for gains in computational efficiency. A body of work contributed by Carolina Osorio, including [15], has focused on

supporting surrogate-based optimization of large-scale urban models with analytical problem specific information.

Our paper contributes to this broader effort by advancing the third branch by introducing a novel metamodel-based equilibration (MBE) scheme. Specifically, we apply a game-theoretical model based on CHANAkYA framework developed in [11] as metamodel to find optimal tolls for an underlying system of directly interfaced **detailed traffic models**. The central hypothesis is that if these faster models can be calibrated effectively, the resulting tolls can perform well when applied to full-scale models.

Our metamodel departs from conventional metamodeling approaches that approximate only the objective function via input-output mapping. Instead, it models the entire transportation system, resulting in a structurally simpler yet functionally equivalent problem at the metamodel level. Further, CHANAkYA framework allows us to model multi-player[1] pricing competitions at the metamodel level. This gives our approach a key advantage over other metamodel-based or Blackbox optimization techniques in general, which typically map a single decision maker's objective function to its decision variables. Thus, they will require independent, sequential metamodel-based optimizations for each decision maker for a multi-player problem. In contrast, our metamodel represents the full multi-player problem, allowing a candidate game-theoretical equilibrium to be computed through a single metamodel-based equilibration. To our knowledge, this is the first instance of solving a multi-player game-theoretical tolling problem in a full-scale detailed traffic model with variable demand.

This paper makes four main contributions:

1) We propose a novel metamodel-based equilibration (MBE) scheme, where a fast CHANAkYA-based model is used to identify desirable tolls for a full-scale traffic simulation.
2) Our method is the first to compute approximate Nash and Stackelberg equilibria in full-scale models with variable demand and multiple decision makers.
3) We introduce a calibration protocol to align the CHANAkYA-based metamodel with the outputs of the detailed traffic model.
4) We demonstrate that well calibrated transport economics and mathematical programming-based models can produce robust tolling strategies in complex, real-world settings.

The rest of the paper is organized as follows: The Methodology section explains the MBE scheme and its components. The Implementation section details the scenario setup. The Results section presents applications to three single decision maker problems and two multi-decision-maker games, showing improved performance over benchmark BO.

## METHODS

The overall metamodel-based equilibration (MBE) scheme is illustrated in **Figure 1**. The term equilibration is used broadly to refer to both game-theoretical equilibrium in multi-player settings and optimization in single decision maker contexts. We represent the detailed traffic model by a directly interfaced travel demand model and full-scale traffic assignment model. The metamodel includes a simpler and more aggregate version of all the relevant stakeholders and subsystems of the underlying setup. In this way, the underlying set-up only has the computational load related to achieving consistency between the directly interfaced travel demand and traffic assignment model, whereas the computational load for the game-theoretical equilibration between multiple decision makers lies completely at the metamodel level. At every iteration, the underlying set-up is used to (re)calibrate the metamodel through a dedicated calibration protocol. The calibrated metamodel then computes new equilibrium tolls, which are transmitted back to the underlying set-up. At the new tolls, the underlying set-up is evaluated again, and the meta-model is recalibrated at the new point. This sequence is repeated until a sufficiently desirable outcome is achieved.

[1] Within the scope of this paper, the term player refers to a decision maker.

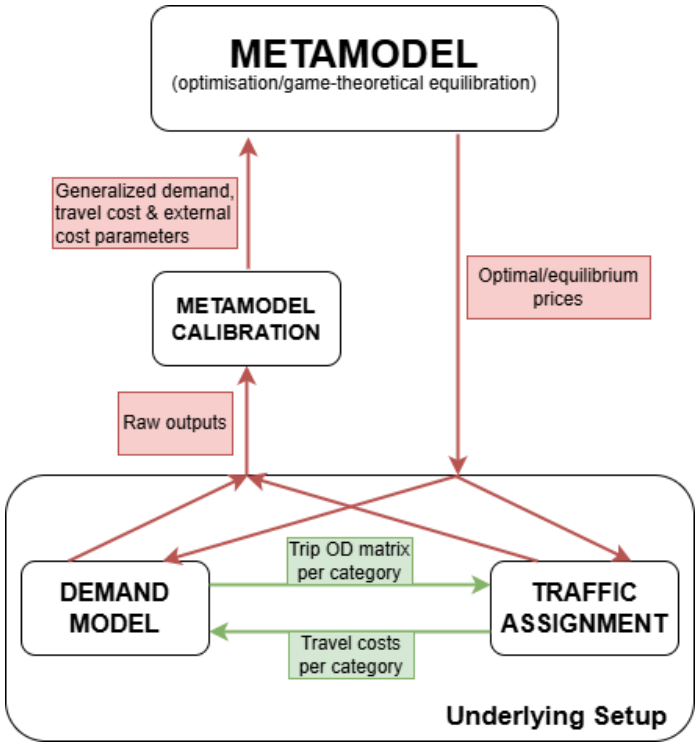



**Figure 1: Illustration of the general metamodel-based equilibration.**

The scheme consists of three main parts: 1) The underlying set-up 2) the metamodel 3) the calibration protocol between the underlying set-up and the metamodel.

**The underlying set-up**

For the scope of this paper, underlying set-up consists of the Static Traffic Assignment Problem (STAP) and an elastic demand model that takes the decision variables i.e., the tolls as input and evaluates their impact on the transportation system.

*Static Traffic Assignment Problem (STAP):*

The STAP is an optimization problem equivalent to Wardrop's equilibrium. It predicts the route-choice of travelers and the resulting flows on the links of the full network. It can be mathematically expressed as:

$$\begin{aligned} \min_{\boldsymbol{X},\boldsymbol{H}} \sum_{(I,J)\in L} \int_0^{X_{IJ}} \left\{ LC_{IJ}(x) + \frac{T_{IJ}}{VoT} \right\} dx \\ X_{IJ} = \sum_{Q \in P} H_Q \Delta_{IJ}^Q \quad \forall (I,J) \in L \\ \text{s.t.} \quad \sum_{Q \in P^{RS}} H_Q = D_{RS} \quad \forall (R,S) \in D \\ H_Q \geq 0 \;\; \forall \, Q \in P \end{aligned} \tag{1}$$

where $L$ is the set of all links $(I,J)$ in the full network $G$, with $I$ being the from-node and $J$ being the to-node. $D$ is the set of all origin-destination pairs $(R,S)$ with $R$ being the origin and $S$ being the destination. $P$ is the set of all paths $Q$ and $P^{RS}$ is the set of all paths between the origin-destination pair $(R,S)$. $X_{IJ}$ and $H_Q$ are flows on link $(I,J) \in L$ and path $Q \in P$ respectively, whereas $D_{RS}$ is the demand

flow between origin-destination pair $(R,S) \in D$. $\Delta_{IJ}^{Q}$ is link-path incidence which denotes whether or not link $(I,J)$ is used by path $Q$. Finally, $LC_{IJ}$ are the link cost functions which are usually represented by the standard BPR function, $T_{IJ}$ are link-based tolls charged on $(I,J) \in L$ and $VoT$ is the monetary value of time in appropriate units. The link-based tolls enter the STAP as exogenously determined constant parameters.

*Elastic demand model:*
We use an elastic demand model (**Equation (2)**) for modelling effects of tolls and changing congestion patterns on the travel demand per OD pair.

$$WTP_{RS} = A_{RS} - B_{RS}.D_{RS} = C_{RS} \quad \forall (R,S) \in D$$
$$\text{OR}$$
$$D_{RS} = (A_{RS} - C_{RS})/B_{RS} \; \forall (R,S) \in D$$
$$C_{RS} = \sum_{(I,J) \in L} LC_{IJ}(x)\Delta_{IJ}^{Q} \tag{2}$$

where $WTP_{RS}$ is the Willingness To Pay (WTP) for a trip between the origin-destination pair $(R,S)$, which is defined by exogenous parameters $A_{RS}$ and $B_{RS}$. $A_{RS}$ signifies the maximum WTP signifying the highest perceived benefit for undertaking a trip and, $B_{RS}$ represents the rate at which the perceived benefit decreases with subsequent trips. $C_{RS}$ is the Wardrop's equilibrium costs between the origin-destination pair $(R,S)$.

The elastic demand model takes $C_{RS}$ as an input from the STAP and determines the demand level $D_{RS}$ for each $(R,S) \in D$. Since STAP in turn takes $D_{RS}$ as an input, there is a mutual dependency between the STAP and the elastic demand model. This mutual dependency is solved as a fixed-point problem using the diagonalization method.

*Impact evaluation:*
Once an acceptable level of mutual consistency is found between the STAP and the elastic demand level, the response of the transportation system to the input tolls is complete and the impact can be quantified. This impact is quantified per decision maker $\mu \in PS$ (set of players) by means of an objective function as in **Equation (3)**.

$$OBJ^{\mu} = -(SBEN^{\mu} - STC^{\mu} + TR^{\mu})$$

where:

$$SBEN^{\mu} = \sum_{(R,S) \in D^{\mu}} \int_{0}^{D_{RS}} (WTP_{RS}) \mathrm{d}D_{RS} = \sum_{(R,S) \in D^{\mu}} \left( A_{RS}.D_{RS} - \frac{B_{RS}.D_{RS}^{2}}{2} \right)$$
$$STC^{\mu} = TTC^{\mu} + SEC^{\mu}$$
$$TTC^{\mu} = \sum_{(R,S) \in D^{\mu}} (C_{RS}.D_{RS})$$
$$SEC^{\mu} = \sum_{(I,J) \in L^{\mu}} \{\Lambda_{(I,J)}.LC_{IJ}\}$$
$$TR^{\mu} = \sum_{(I,J) \in L^{\mu}} \left[ \left\{ \sum_{T \in T_{control}^{\mu}{}_{(I,J)}} \frac{T_{IJ}}{VoT} \right\}.X_{(I,J)} \right] \tag{3}$$

$D^{\mu}$ is the set of OD pairs under the constituency of decision maker $\mu$. $SBEN^{\mu}$ and $STC^{\mu}$ represent, respectively, the travel benefit and total costs due to travel of the society under the constituency of decision maker $\mu$. $TTC^{\mu}$ and $SEC^{\mu}$ represent the travel costs and the social external costs (e.g., from

accidents, pollution etc.) of the society under jurisdiction of decision maker $\mu$, respectively. The latter are costs that users don't internalize while making their travel decisions. $\Lambda_{(I,J)}$ is pre-decided scalar factor indicating the ratio of link $(I,J)$ 's social external cost to its travel time and $L^{\mu}$ is the set of links under the jurisdiction of government type decision maker $\mu$ and $TR^{\mu}$ is the revenue generated from these links. $T_{control}{}^{\mu}_{(I,J)}$ is the set of tolls controlled by decision maker $\mu$ on link $(I,J)$.

All the terms in $OBJ_{\mu}$ are in units of time and a lower value of $OBJ_{\mu}$ is more desirable.

**The metamodel**

The metamodel is a game-theoretical model of pricing in transportation systems based on the CHANAkYA framework [10], [11]. At the core of the metamodel lies a Mixed Integer Quadratic Program (MIQP) that serves as the problem of a single decision maker trying to find optimal tolls for an underlying demand and route-choice user-equilibrium (UE). This problem is expressed as **Equation (4)**.

The crucial difference between the underlying set-up and the UE in metamodel is that the transportation system is modelled in a simplified and aggregate way at the metamodel level. On the supply side, a spatially abstract network $g$ with simplified link cost functions is used instead of the full physical network $G$ with the standard BPR link cost functions. Similarly on the demand side, only aggregate origin-destination pairs are modelled, which are highly relevant for the desired analysis. The purpose is to model only as much detail as needed to distinguish between traveler types and between their travel options that are subject to different tolls being optimized or have a different nature.

With these simplifications and aggregations, the metamodel UE problem becomes computationally much easier to solve than that in the underlying setup.

$$\min_{\boldsymbol{h},\boldsymbol{d},\boldsymbol{\tau^{\mu}}} obj^{\mu}$$

$$\text{s.t. } UE$$

where $UE$:

$$\begin{gathered}
\sum_{q \in p_{rs}} h_q = d_{rs} \; \forall (r,s) \in d \\
h_q \geq 0 \;\; \forall q \in p \\
d_{rs} \leq d_max_{rs} \; \forall (r,s) \in d \\
h_q - M.I_q \leq 0 \;\; \forall q \in p \\
M.\left(wtp_{rs} - pc_q\right) + \left(1 - I_q\right) \leq 0 \;\; \forall q \in p_{rs}, \qquad \forall (r,s) \in d \\
-\left(wtp_{rs} - pc_q\right) + M.I_q - M \leq 0 \;\; \forall q \in p_{rs}, \qquad \forall (r,s) \in d
\end{gathered} \tag{4}$$

$M$ is a large but finite constant. $I_q$ are binary variables for all paths $q \in P$ and are used to denote whether a path is active or inactive. $d_max_{rs}$ is an upper bound on $d_{rs}$ for all $(r,s) \in d$. $pc_q$ are path costs for all paths $q \in p$. These can be derived from link costs ($lc_{ij}$) using **Equation (5)**. $l$ is the set of all links of the simplified abstract network $g$ and $\tau_{ij}$ are the link-based tolls applied on link $(i,j) \in l$. It should be noted that unlike in the underlying setup where tolls were exogenously determined, in the single-player optimization problem of **Equation (4)**, the tolls ($\boldsymbol{\tau^{\mu}}$) controlled by decision maker $\mu$ are endogenous optimization variables. All other terms and variables are completely analogous to their uppercase counterparts in the underlying setup.

$$\begin{gathered}
pc_q = \sum_{(i,j) \in l} \left[\delta_{ij}^{q}.\left(lc_{ij}(x) + \frac{\tau_{ij}}{VoT}\right)\right] \\
lc_{ij} = \alpha_{ij} + \beta_{ij}.x_{ij} \;\; \forall (i,j) \in l \\
wtp_{rs} = a_{rs} - b_{rs}.d_{rs} \;\; \forall (r,s) \in d
\end{gathered} \tag{5}$$

$$obj^{\mu} = -(sben^{\mu} - stc^{\mu} + tr^{\mu}) \tag{6}$$

where:

$$sben^{\mu} = \sum_{(r,s) \in d^{\mu}} \int_{0}^{d_{rs}} (wtp_{rs}) \mathrm{d}d_{rs} = \sum_{(r,s) \in d^{\mu}} \left( a_{rs}.d_{rs} - \frac{b_{rs}.d_{rs}^{2}}{2} \right)$$

$$stc^{\mu} = ttc^{\mu} + sec^{\mu}$$

$$ttc^{\mu} = \sum_{(r,s) \in d^{\mu}} \left( \sum_{q \in p_{rs}} h_q.(pc_q) \right)$$

$$sec^{\mu} = \sum_{(i,j) \in l^{\mu}} \{\lambda_{(i,j)}.lc_{ij}\}$$

$$tr^{\mu} = \sum_{(i,j) \in l^{\mu}} \left[ \left\{ \sum_{\tau \in \tau_{control}{}^{\mu}_{(i,j)}} \frac{\tau_{ij}}{VoT} \right\} . x_{(i,j)} \right] \tag{7}$$

The multi-player problems in the metamodel are built upon the single-player optimization. When all the involved decision makers are making simultaneous decisions without anticipating reactions from each other, the resulting game is a Nash or Nash-Cournot game. Despite the simplifications in the metamodel UE, such a game remains a theoretically complex Nonlinear Generalized Nash Equilibrium Problem (NGNEP). It is modelled as a fixed-point problem and is solved using the diagonalization method as shown in **Figure 2**. $PS$ is the set of decision makers.

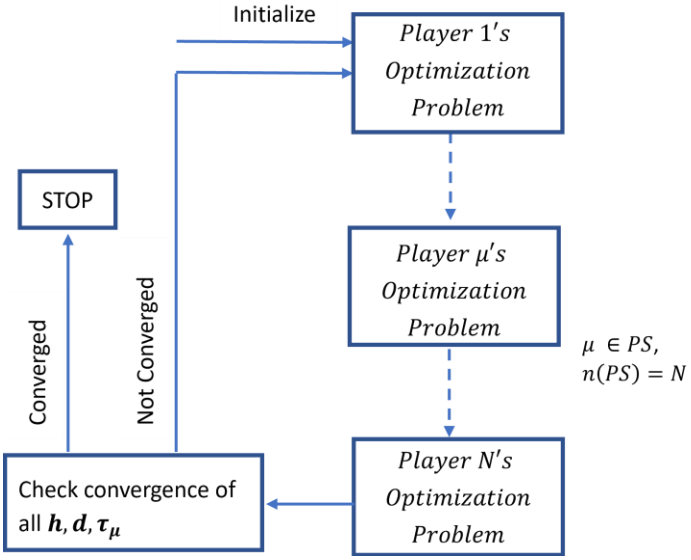



**Figure 2: Representation of Nash (Nash-Cournot) game solution process.**

If there is a decision-maker which anticipates the response of followers, the resulting game is a Stackelberg game. This becomes an optimization problem subject to the single follower's MIQP in case of a single follower or the NGNEP of multiple followers in case of multiple followers. This is depicted in **Figure 3**. $\mu_l$ is the decision maker acting as the Stackelberg leader and $PS_f$ is the set of follower decision makers $\mu_f$. Such problems are highly non-convex and are not suitable to be solved by exact methods. These will be solved using Bayesian Optimization.

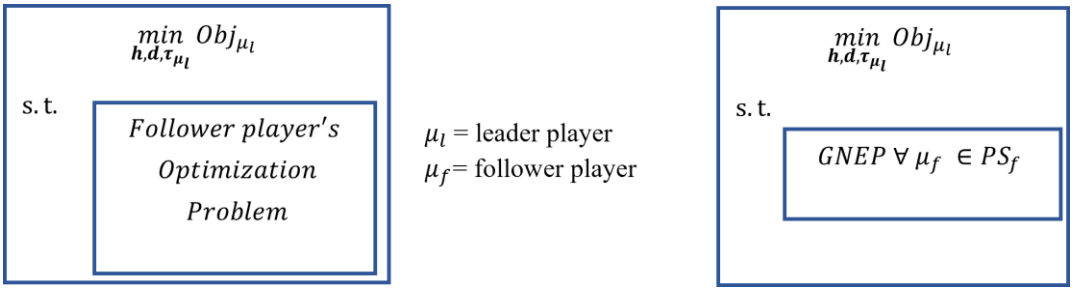



**Figure 3: Representation of Stackelberg game as an optimization problem of the leader with single follower (left) and followers in a Nash game (right).**

The core methodology and the general framework of the metamodel is covered in detail in [10], [11]. In this paper we reserve ourselves to a specific instance of the framework to showcase its utility as a meta or surrogate model. This type of application is completely novel not only for the CHANAkYA framework but also for other similar types of models.

**The calibration protocol**

For the optimal tolls identified through metamodel-level optimization to result in desirable outcomes at the underlying setup, it is essential that the simplified user equilibrium (UE) in the metamodel serves as a valid approximation of the underlying setup. This alignment is ensured by calibrating metamodel parameters. These include the demand side parameters $a_{rs}$ and $b_{rs}$, and the supply side parameters $\alpha_{ij}$, $\beta_{ij}$ and $\lambda_{(i,j)}$ which must be determined before solving the optimization problem of **Equation (4)**.

The calibration protocol is divided into two steps, one for the demand side parameters and one for the supply side parameters.

*Demand side parameters:*

The demand side parameters $\boldsymbol{a}$ and $\boldsymbol{b}$ are specified analytically using an aggregation procedure.[2] The zeroth step in this process involves classifying the detailed OD pairs $(R,S)\ \in\ D$ of the underlying setup into OD pairs of the metamodel level $(r,s)\ \in\ d$. Each $(R,S)\ \in\ D$ may be mapped to at most one $(r,s)\ \in\ d$, or remain unclassified if no appropriate mapping is found. This leads to a mapping matrix $[\theta]$ of size $n(d)$ X $n(D)$, where $\theta_{rs,RS} = 1$ if the detailed OD pair $(R,S)$ is mapped to the metamodel OD pair $(r,s)$, and 0 otherwise.

Two key results are required from the underlying setup i.e., demand vector $\boldsymbol{D_{US}}$ and travel costs vector $\boldsymbol{C_{US}}$ which are mutually consistent with each other.[3] These are available after the elastic demand model and the STAP from the underlying setup have arrived at a fixed point. Three other vectors are also required:

1) Reference OD vector $\boldsymbol{D_{US}^{ref}}$: base demand at the reference tolls. This is usually available from the regional or national mobility departments.
2) Reference travel costs vector $\boldsymbol{C_{US}^{ref}}$: obtained by assigning the reference OD vector $\boldsymbol{D_{US}^{ref}}$ on the real physical network using STAP.
3) Maximum demand OD vector $\boldsymbol{D_{US}^{max}}$: an upper bound on OD demand, chosen based on the scope and objectives of the analysis.

Given the mapping matrix $[\theta]$ and these results, $\boldsymbol{a}$ and $\boldsymbol{b}$ can be calculated in closed form, as detailed in **Equation (8)**. **Equation (8)** aggregates the demand $\boldsymbol{D_{US}}$ vector from the underlying setup into

[2] The bold signifies vector of parameters for all $(r,s)\ \in\ d$.
[3] The bold and the subscript US signifies vector of all $(R,S)\ \in\ D$.

an equivalent demand $\boldsymbol{D_m}$ for the metamodel OD pairs. **Equation (8)** calculates average travel costs for the metamodel OD pairs weighted by the demand $\boldsymbol{D_{US}^{max}}$. **Equations (8)- (8)** perform the same operations on the reference demand $\boldsymbol{D_{US}^{ref}}$ and reference travel costs $\boldsymbol{C_{US}^{ref}}$, yielding $\boldsymbol{D_m^{ref}}$ and $\boldsymbol{C_m^{ref}}$ respectively**.** This gives two points, each representing a mutually consistent pair of demand and travel costs, and $\boldsymbol{a}$ and $\boldsymbol{b}$ can be calculated by fitting a linear curve on these points using **Equations (8)- (8)**.

$$\boldsymbol{D_m} = [\theta] \text{ X } \boldsymbol{D_{US}} \tag{8.1}$$

$$\boldsymbol{C_m} = \frac{\left([\theta] \text{ X } (\boldsymbol{D_{US}^{max}}.\boldsymbol{C_{US}})\right)}{([\theta] \text{ X } \boldsymbol{D_{US}^{max}})} \tag{8.2}$$

$$\boldsymbol{D_m^{ref}} = [\theta] \text{ X } \boldsymbol{D_{US}^{ref}} \tag{8.3}$$

$$\boldsymbol{C_m^{ref}} = \frac{\left([\theta] \text{ X } \left(\boldsymbol{D_{US}^{max}}.\boldsymbol{C_{US}^{ref}}\right)\right)}{([\theta] \text{ X } \boldsymbol{D_{US}^{max}})} \tag{8.4}$$

$$\boldsymbol{b} = -\frac{\left(\boldsymbol{C_m^{ref}} - \boldsymbol{C_m}\right)}{\left(\boldsymbol{D_m^{ref}} - \boldsymbol{D_m}\right)} \tag{8.5}$$

$$\boldsymbol{a} = \boldsymbol{C_m} + \boldsymbol{b} * \boldsymbol{D_m^{ref}} \tag{8.6}$$

(8)

*Supply side parameters:*

To determine the values of the metamodel link cost parameters i.e., $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$ and $\boldsymbol{\lambda}$, we formulate an calibration (optimization) problem (see **Equation (9)**) that minimizes the mismatch between the toll sensitivities computed at the underlying setup and those computed at the metamodel level for various components of the decision maker's objective function (see **Equation**s **(3)** and **(7)**).

$$\min_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\lambda}} \sum_{s \in S} (F_s . Mismatch(s))$$

where:

$$Mismatch(s) = (s_{US} - s_{meta}(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\lambda}))^2 \ \forall \ s \in S$$
$$S = \{Sens_{Obj}, Sens_{Sben}\ Sens_{Ttc}, Sens_{Sec}, Sens_{Tr}\}$$

(9)

$s_{US}$ is the value associated with the uppercase (underlying setup) version of $s \in S$, and $s_{meta}$ is the value associated with the lowercase (metamodel) version of $s \in S$, e.g.,

if,

$$s = Sens_{Obj} \in S$$

then,

$$s_{US} = Sens_{OBJ}$$
$$s_{meta} = Sens_{obj}$$
$$Mismatch(Sens_{Obj}) = \left(Sens_{OBJ} - Sens_{obj}(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\lambda})\right)^2$$

(10)

Note that $s_{meta}$ is dependent on the calibration parameters $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$ and $\boldsymbol{\lambda}$ whereas $s_{US}$ is exogenously defined and is computed using the raw inputs from the underlying setup. $F_s$ is a scalar factor that helps in adjusting the relative importance of the mismatches in sensitivities of different terms.

Calculating the different toll sensitivities at the underlying setup is a crucial concern. The underlying setup is a non-analytical simulation which rules out analytically deriving these sensitivities. Underlying setup is also supposed to be an expensive simulation which rules out targeted numerical computation of these sensitivities. Thus, we compute these sensitivities exploiting the simulation history of the underlying setup. We find the tolls ($\boldsymbol{\tau^{\mu}}$) in the history which are closest to the current tolls and compute the $s_{US}$ as in **Equation (11)** for $s = Sens_{Obj}$.

$$Sens_{OBJ} = 2 * \frac{(OBJ_{closest} - OBJ)}{abs(OBJ_{closest} + OBJ)} \tag{11}$$

The $s_{meta}$ are also computed using a similar formula e.g., for $s = Sens_{Obj}$ in **Equation (12)**. However, this computation cannot rely on historical data. This is because both $obj$ and $obj_{closest}$ depend on $\boldsymbol{a}$, $\boldsymbol{b}$, $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$ and $\boldsymbol{\lambda}$. $\boldsymbol{a}$ and $\boldsymbol{b}$ have only been identified in the step 1 of the calibration protocol whereas $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$ and $\boldsymbol{\lambda}$ are completely endogenous to the calibration problem. Thus, $s_{meta}$ needs to be computed endogenously for each candidate value of $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$ and $\boldsymbol{\lambda}$ within the calibration problem. This is done numerically by simulating metamodel UE at the current and the closest tolls for each candidate value of $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$ and $\boldsymbol{\lambda}$, and the current values of $\boldsymbol{a}$ and $\boldsymbol{b}$ identified in step 1. Since metamodel UE is computationally inexpensive, the overall computational burden remains quite low.

$$Sens_{obj} = 2 * \frac{(obj_{closest} - obj)}{abs(obj_{closest} + obj)} \tag{12}$$

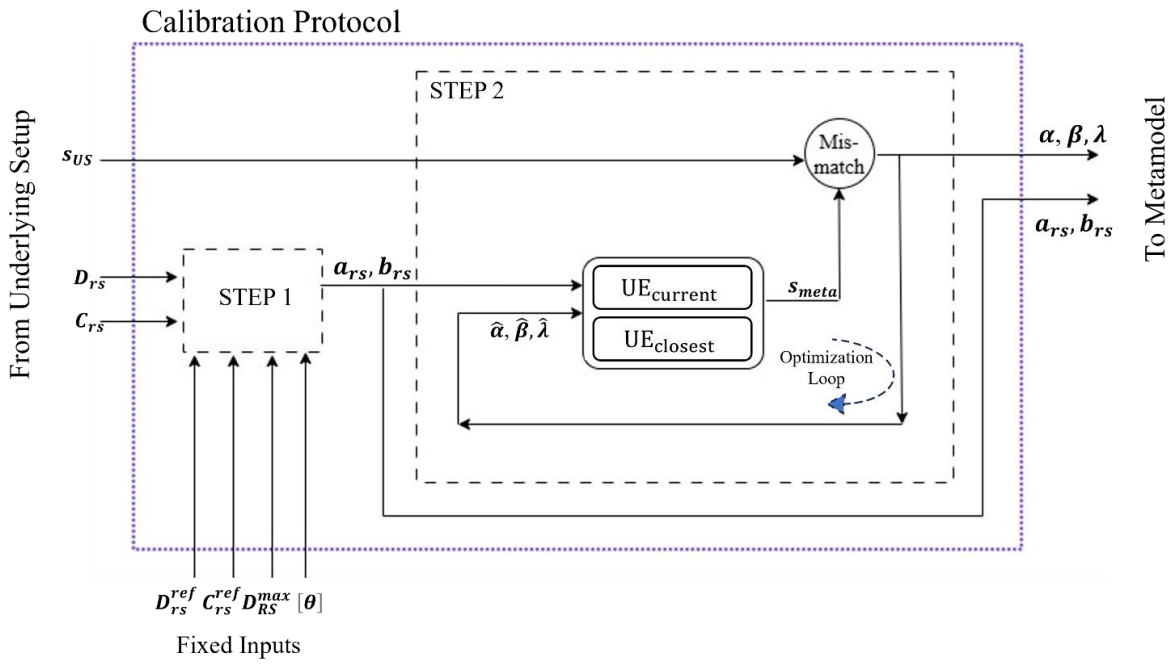



**Figure 4: Schematic view of the calibration protocol including Step 1 (for demand parameters) and Step 2 (for supply parameters).**

The calibration problem is quite non-convex owing to internal computation of sensitivities and is solved by Bayesian Optimization.

The overall scheme of the entire calibration protocol including the calibration problem of step 2 can be seen in block diagram shown in **Figure 4.** With detailed explanations of the underlying setup, the metamodel and the calibration protocol, the general scheme of MBE approach (**Figure 1**) can be further specified as shown in **Figure 5**.

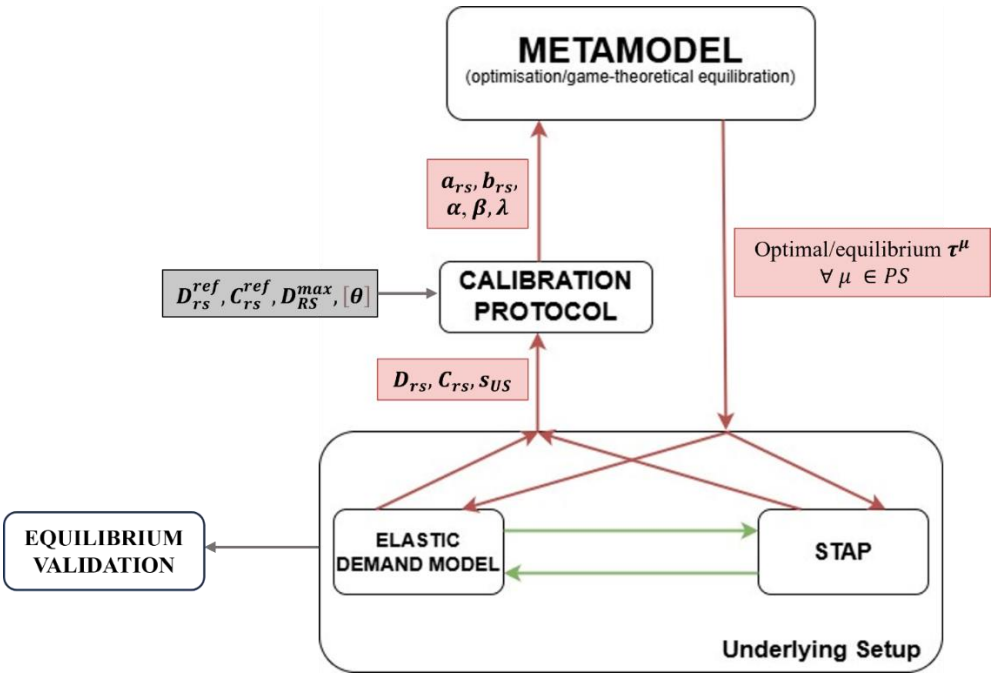



**Figure 5: Illustration of the metamodel-based equilibration scheme with a specified calibration protocol**.

**Equilibrium validation**

Once the calibrated metamodel has identified equilibrium tolls, it remains necessary to evaluate whether these tolls constitute a valid equilibrium at the level of the underlying setup. This validation procedure is referred to as equilibrium validation in **Figure 5**.

For a single decision maker problem, the equilibrium validation, or in this case desirability evaluation, is relatively straight-forward. The tolls which lead to a lower value of $OBJ^{\mu}$ are more desirable.

However, when multiple decision makers are involved, each optimizing a separate set of tolls, the goal shifts from finding optimal tolls to identifying a game-theoretical equilibrium. The tolls which lead to the lowest $OBJ^{\mu}$ for a certain decision maker may not correspond to a valid equilibrium. We cover Nash and Stackelberg games with each requiring a slightly different equilibrium validation procedure.

*Nash game:*

Before describing the equilibrium validation procedure, we clarify key concepts. A Nash Equilibrium (NE) is a state where no decision maker can improve their outcome by unilaterally changing their tolls, assuming others' tolls are fixed. A local NE relaxes this by requiring that no decision maker can benefit from an *infinitesimal* change to their tolls. An ε-local NE allows for a small improvement ($\leq \varepsilon$) under such infinitesimal changes. Extending this numerically, a numerical ε-local NE is one where no decision maker can improve their objective by more than ε through a small but finite change ($\phi$), while others' tolls remain fixed.

Given this terminology, the NE validation proceeds as follows. After executing the underlying setup at the equilibrium tolls $\boldsymbol{\tau^{\mu}}$ for all $\mu \in PS$ identified by the metamodel, we compute $\varepsilon_{\mu+}$ and $\varepsilon_{\mu-}$ for

all $\mu \in PS$. $\varepsilon_{\mu+}$ denotes a change in the $OBJ^{\mu}$ for an increase in the decision maker $\mu$'s tolls by $\phi$ with all other tolls being fixed, whereas $\varepsilon_{\mu-}$ does the same for a decrease in decision maker $\mu$'s tolls by $\phi$.

$$\varepsilon_{\mu+} = OBJ^{\mu}\left(\boldsymbol{\tau}^{\boldsymbol{\mu}}, \boldsymbol{\tau}^{\{\boldsymbol{PS}\}-\{\boldsymbol{\mu}\}}\right) - OBJ^{\mu}\left(\boldsymbol{\tau}^{\boldsymbol{\mu}} + \phi, \boldsymbol{\tau}^{\{\boldsymbol{PS}\}-\{\boldsymbol{\mu}\}}\right)$$
$$\varepsilon_{\mu-} = OBJ^{\mu}\left(\boldsymbol{\tau}^{\boldsymbol{\mu}}, \boldsymbol{\tau}^{\{\boldsymbol{PS}\}-\{\boldsymbol{\mu}\}}\right) - OBJ^{\mu}\left(\boldsymbol{\tau}^{\boldsymbol{\mu}} - \phi, \boldsymbol{\tau}^{\{\boldsymbol{PS}\}-\{\boldsymbol{\mu}\}}\right)$$
$$\forall\, \mu \in PS \tag{13}$$

If for a particular iteration of the MBE scheme, $\varepsilon_{\mu+} \leq \varepsilon$ and $\varepsilon_{\mu-} \leq \varepsilon$ for all $\mu \in PS$, then the corresponding tolls qualify as a valid numerical ε-local NE. When multiple iterations satisfy numerical ε-local NE conditions, they may be ranked based on the average value of $\varepsilon_{\mu+}$ and $\varepsilon_{\mu-}$(including only positive values) over all $\mu \in PS$. The tolls where this average is closest to zero are more valid candidates for Nash equilibria.

*Stackelberg game:*

Unlike in a Nash game, where each decision maker assumes the tolls of others to be fixed, the Stackelberg leader internalizes the impact of its tolls on the subsequent tolls of the followers. This hierarchical structure increases the computational and conceptual complexity of the validation of the equilibrium tolls identified by the metamodel.

Nevertheless, the validation of the response of followers remains the same as in Nash game. It involves computing $\varepsilon_{\mu+}$ and $\varepsilon_{\mu-}$ for all $\mu_f \in PS_f$. If $\varepsilon_{\mu+} \leq 0$ and $\varepsilon_{\mu-} \leq 0$ (or both $\leq \varepsilon$) for all $\mu_f \in PS_f$, then the followers have no (or very small) incentive to deviate from their tolls for the given tolls of the Stackelberg leader and their responses are considered valid. In contrast, verifying the validity of the Stackelberg leader's tolls given a valid response of the followers is much more complicated. This is because the valid response of the followers may itself change when the Stackelberg leader's tolls are perturbed from $\boldsymbol{\tau}^{\boldsymbol{\mu_l}}$ to $\boldsymbol{\tau}^{\boldsymbol{\mu_l}} \pm \phi$. Validating this is computationally infeasible. However, we use a workaround. Once a set of iterations that yield a valid followers' response have been identified, the tolls which lead to a lower $OBJ^{\boldsymbol{\mu_l}}$ denote a more valid Stackelberg equilibrium (SE).

**Benchmark**

To evaluate the performance of our novel MBE scheme, a benchmark is needed. We distinguish between the single-player and multi-player problem settings.

*Single-player problem:*

Even for a single decision maker, each evaluation of the objective function $OBJ^{\mu}$ requires a full run of the expensive underlying setup, making the problem complex, non-convex, and non-analytical. Such problems are typically tackled using general-purpose metamodel-based methods like BO. This requires many evaluations to explore and exploit the decision space effectively. If $t_{US}$ is the time per evaluation and $n_{US}$ the required evaluations, the total computational cost is $n_{US}.t_{US}$ hours. For benchmarking, we compare our MBE scheme to BO both in terms of performance under matched computational budgets and solution quality (achieved by using BO with higher computation budget).

*Multi-player problem(s):*

Benchmarking multi-player problems is significantly more complex. Unlike standard metamodels (e.g., in BO), the metamodel in our MBE scheme approximates the entire transport system, allowing game-theoretical equilibrium computation in multi-player problems. In contrast, BO requires separate optimizations per decision maker for this.

In a Nash game with $n_{PS}$ decision makers and $n_N$ fixed-point iterations each involving $n_{US}$ evaluations of the underlying setup, the total computational budget becomes $n_N.n_{PS}.n_{US}.t_{US}$ hours. Stackelberg games are worse: with one follower, the leader must run $n_S$ nested BOs, costing $n_S.n_{US}.t_{US}$ hours; with multiple followers, the cost becomes $n_S.n_N.n_{PS}.n_{US}.t_{US}$ hours.

Given a typical $t_{US}$ = 0.5-2 hours, and $n_S$, $n_N$ and $n_{US}$ range from 20 to 100, benchmarking in multi-player problems becomes computationally infeasible, underscoring the need for specialized schemes like MBE.

## IMPLEMENTATION

We apply the proposed metamodel-based equilibration (MBE) scheme to a case study in Aarschot, Belgium. The city's ring road is neither circular nor fully closed (see **Figure 6**), so travelers choose among three main route types: (i) radial roads through the city center, (ii) the fragmented ring road, or (iii) local roads via surrounding municipalities. To influence route choices and reduce congestion, we optimize three tolls: $t_{rad}$ to enter radial roads, $t_{ring}$ to enter the city's ring road and $t_m$ to enter the local roads of surrounding municipalities.




**Figure 6: The city of Aarschot with an open and asymmetrical ring road.**

### The underlying set-up

*Static Traffic Assignment Problem (STAP):*

The STAP requires the specification of four key components: (i) a detailed road network with a sufficient buffer around the city, (ii) a zonal system with a matching buffer, (iii) a corresponding OD matrix, and (iv) a solver.

Using an in-house method, we extracted the road network for Aarschot with a buffer radius of 17 km (**Figure 7**) and derived a 173-zone system from a national zoning dataset provided by the Departement Mobiliteit en Openbare Werken (MOW) (**Figure 8**).

An OD matrix for the morning peak (1 hour) was extracted from the MOW-provided national matrix. External and through-traffic were heuristically assigned to peripheral zones, yielding a reference OD matrix and flattened demand vector $\boldsymbol{D}_{\boldsymbol{RS}}^{\boldsymbol{ref}}$.

We chose PTV Visum to solve the STAP. The reference cost matrix and the reference cost vector ($\boldsymbol{C}_{\boldsymbol{RS}}^{\boldsymbol{ref}}$) were obtained by performing an initial STAP with reference OD matrix as the demand.

To implement the three tolls $t_{rad}$, $t_{ring}$ and $t_m$, the study area was divided into Territory 5 (surrounding municipalities) and Territory 6 (city center), as shown in **Figure 9**. Tolling rules were defined as follows:

1. Radial toll, $t_{rad}$ was applied to:
   - All links entering Territory 6 from outside except when using the primary roads (shown in yellow and orange in **Figure 9**).
   - All off-ramps from the primary roads to the local roads inside Territory 6.
2. Ring toll, $t_{ring}$ was applied to:

- All primary roads entering Territory 6.
- All on-ramps to primary roads originating in Territory 6.

3. Municipal toll, $t_m$ was applied to:
   - All links entering Territory 5, either from inside or outside, except when using the primary roads.
   - All off-ramps from the primary roads to the local roads inside Territory 5.

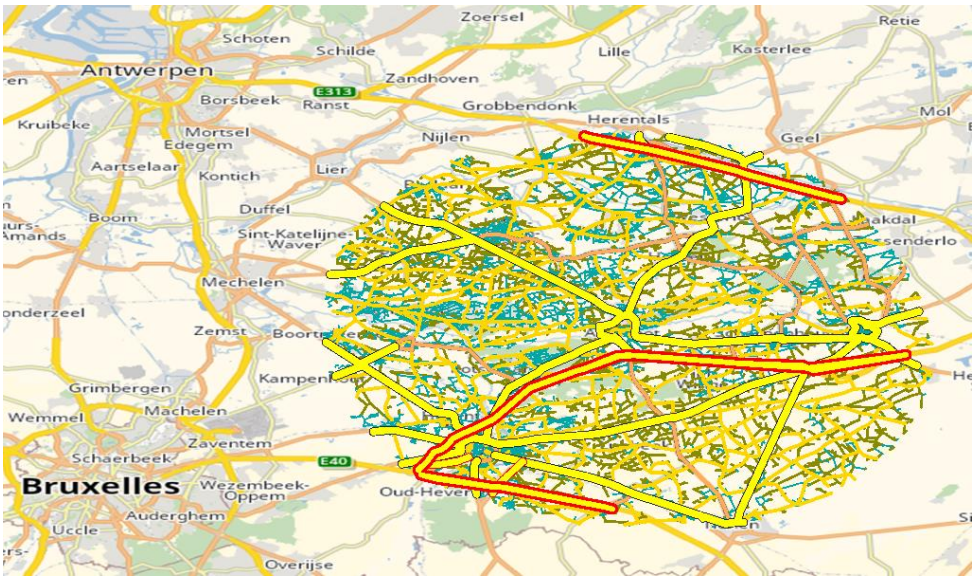



**Figure 7: Aarschot network shown against the background of OpenStreetMap.**

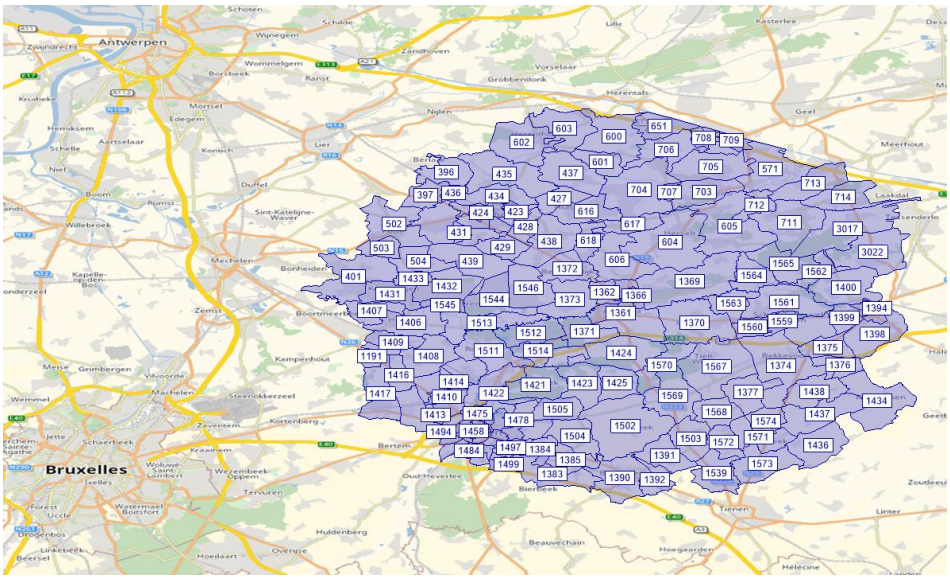



**Figure 8: 173 zones used for the study.**

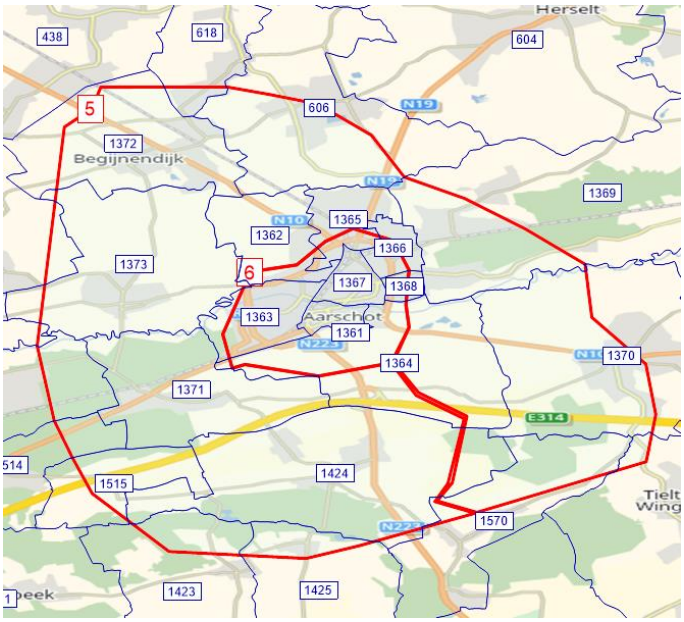



**Figure 9: Territory 6 is under city jurisdiction and Territory 5 is under jurisdiction of surrounding rural municipalities.**

*Elastic demand model:*

To specify the inverse demand (WTP) function in **Equation (2)**, vectors $A_{RS}$ and $B_{RS}$ were estimated for all 173.173 = 29929 OD pairs using: (i) the reference OD vector ($\boldsymbol{D_{RS}^{ref}}$**),** (ii) reference cost vector ($\boldsymbol{C_{RS}^{ref}}$), (iii) a realistic assumption about the maximum possible demand ($\boldsymbol{D_{RS}^{max}}$). This yields two cost-demand points for each of the 29929 OD pairs: one at reference conditions and one at zero cost. These two cost-demand points per OD pair were used to linearly approximate $A_{RS}$ and $B_{RS}$. The horizontal interaction between the STAP and the elastic demand model (**Figure 1Error! Reference source not found.** and **Figure 5**) was solved as a fixed-point problem with Method of Successive Averages (MSA) smoothening.

*Impact Evaluation:*

Depending on the scenario, tolls are controlled by either one or two decision makers (city and/or surrounding municipalities). All tolls are expressed in time units (VoT = 1). Links are categorized to assign social external cost ratios $\Lambda_{(I,J)}$:

1. Radial links (non-primary links in territory 6): $\Lambda_{RAD} = 2$ (unless otherwise specified).
2. Ring links (primary links in territory 6): $\Lambda_{RING} = 0$.
3. Surrounding municipality links (non-primary links in territory 5): $\Lambda_{M} = 0.5$ (unless otherwise specified).

Using these values, $OBJ^{\mu}$ was evaluated for each $\mu \in PS$ using **Equation (3)**.

**The metamodel**

The MBE metamodel requires only: (i) the network topology and (ii) the OD pair distribution. All other parameters (calibration protocol) and variables (metamodel problem) are internal to the MBE scheme. We use a 13-link "sling-bag" network (**Figure 10**) to represent the full network.

- Radial roads: Links $l5$ and $l6$ (entry via link $l5$ tolled with $t_{rad}$).
- Ring road: Links $l7$, $l4$ and $l8$ (entry via Link $l7$ tolled with $t_{ring}$).
- Municipal roads: Links $l9$, $l10$, $l3$, $l11$ and $l12$ (entry via links $l10$ and $l12$, tolled with $t_m$)
- External bypass: Link $l13$, untolled.

On the demand-side, 10 OD pairs $(r, s) \in d$ are modelled (**Figure 10** caption).

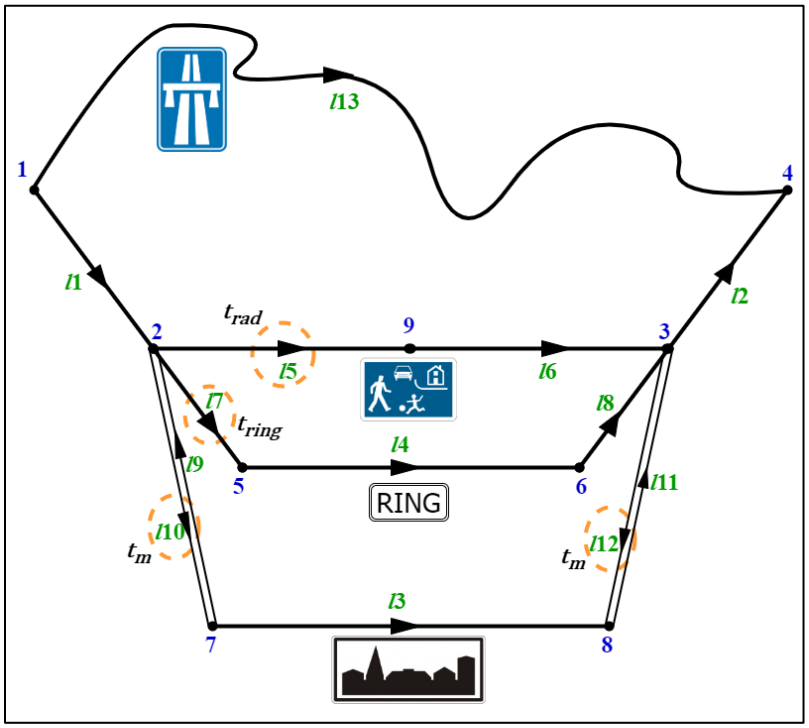



**Figure 10: Network for metamodel: Node ID (blue), link indices (green) and circled link indices of tolled links. d = {(1,4),(1,7),(1,8),(7,4),(8,4),(1,9),(9,4),(7,8),(7,9),(9,8)}.**

**The calibration protocol**

Demand-side calibration requires a mapping matrix $[\theta]$ of size 10 X 29929 to link metamodel OD pairs to those in the underlying setup. Due to its size, this matrix is not reproduced.

Supply-side calibration requires scaling factors $F_s$ each sensitivity component $s \in \{Sens_{Obj}, Sens_{Sben}, Sens_{Ttc}, Sens_{Sec}, Sens_{Tr}\}$. We used $F_s = 10$ for $Sens_{Obj}$, 2 for $Sens_{Tr}$ and 1 for the rest.

**RESULTS**

We apply the proposed MBE scheme to the Aarschot toll-setting case, starting with single-player problems and later extending to multi-player settings.

**Single-player problem**

A unified decision maker optimizes all three tolls: $t_{rad}$, $t_{ring}$ and $t_m$. The MBE scheme is initialized with tolls of [0.25, 0.25, 0.25] hours and bounds of [0, 1] hour. We apply Bayesian Optimization (BO) with 30 iterations to have a near global optimum for benchmarking the quality of the solution. Then we apply our MBE scheme with 8 iterations. Additionally, we perform a BO with 10 iterations to compare the MBE scheme's performance with BO under similar computational budget.

*No external costs*

We assumed zero external costs at the underlying setup i.e., $\Lambda_{RAD} = \Lambda_{RING} = \Lambda_M = 0$ to isolate congestion impacts. The evolution of $OBJ^{\mu}$ is shown in **Figure 11** for BO with 30 iterations. It identified best $OBJ^{\mu}$ value of -934.77 vehicle-hours at toll values of [0.01, 0, 0] hours in 11 hours of computation. These low values of tolls are intuitive, as the city of Aarschot experiences minimal congestion. Further, in

the absence of external costs, any significant increase in travel costs via tolls is likely to reduce the travel benefit of the society ($SBEN^{\mu}$) more strongly than it reduces the social costs of travel ($STC^{\mu}$).

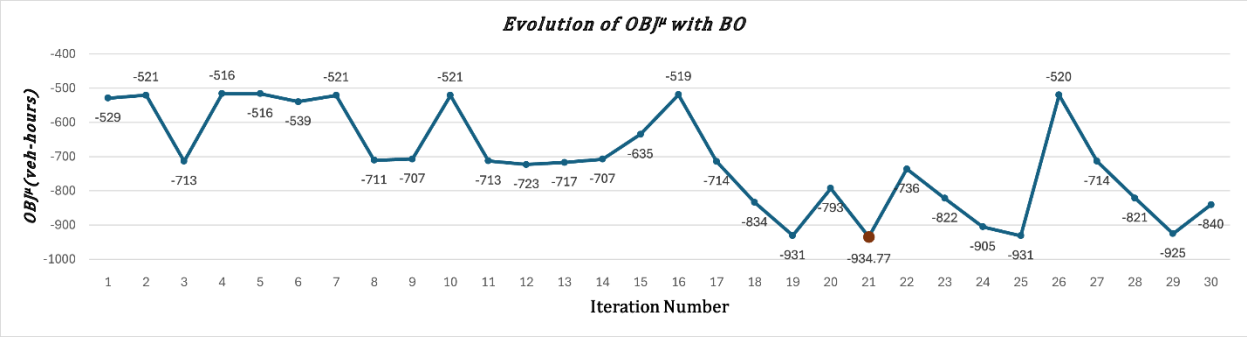



**Figure 11: Bayesian Optimization (BO) applied to single-player problem with no external costs.**

The evolution of $OBJ^{\mu}$ and tolls with our MBE scheme are shown in **Figure 12** and **Figure 13** respectively. The 8 iterations with MBE took 4.49 hours including 26.3 minutes of preprocessing. It can be seen in **Figure 12** that $OBJ^{\mu}$ declines nearly monotonously over the first 6 iterations. Moreover, the best solution, identified in iteration 6, has an $OBJ^{\mu}$ value of -939.14 vehicle-hours which not only matches the benchmark of -934.77 vehicle-hours but in fact goes further beyond it. This best $OBJ^{\mu}$ value corresponds to the toll values of [0, 0, 0.007] hours. Finally, the evolution of $OBJ^{\mu}$ is shown in **Figure 14** for BO with 10 iterations. It identified a best $OBJ^{\mu}$ value of only -713.49 vehicle-hours, despite a comparable computation time of 4.29 hours.

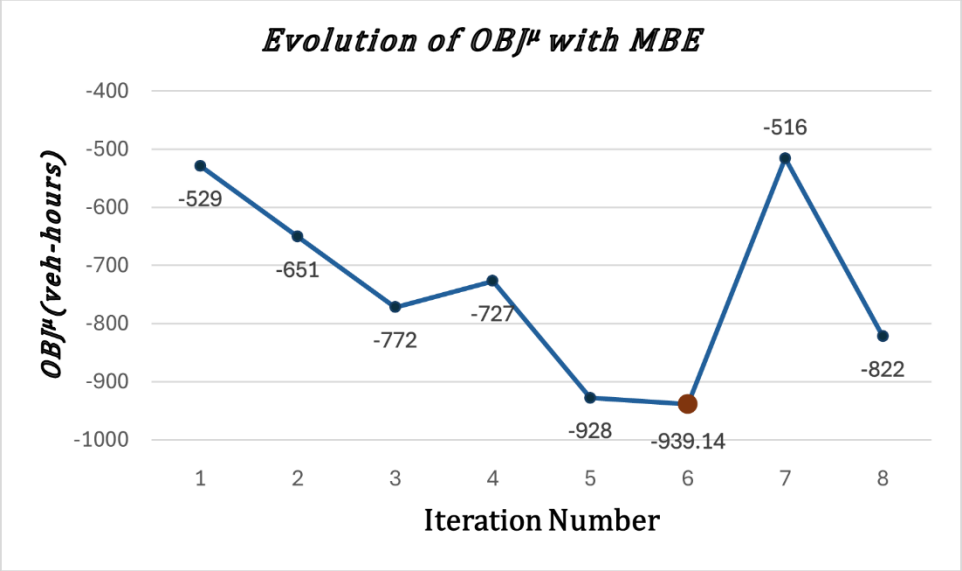



**Figure 12: Metamodel-based equilibration (MBE) scheme applied to single-player problem with no external costs.**

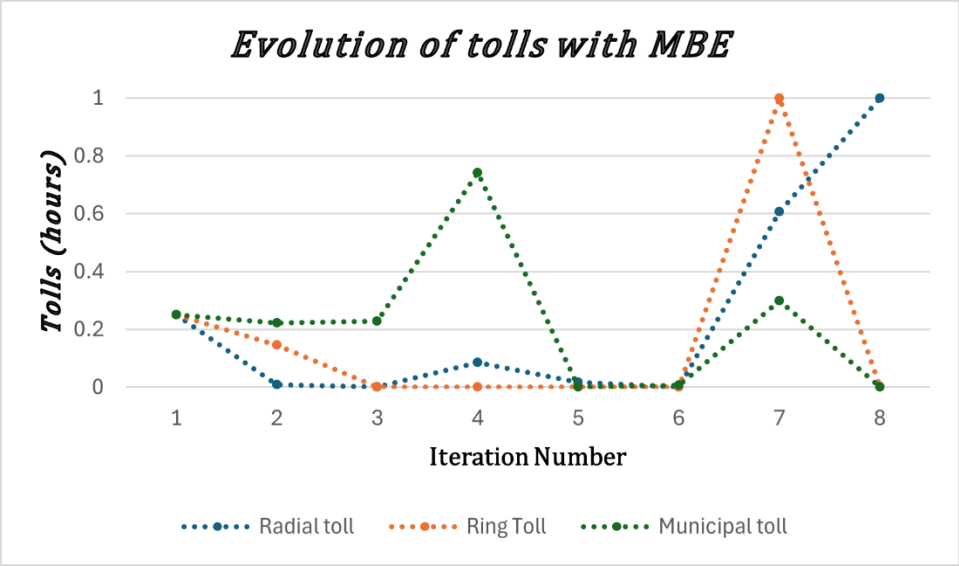



**Figure 13: Metamodel-based equilibration (MBE) scheme applied to single-player problem with no external costs.**

The MBE scheme not only surpassed BO under comparable computational budgets but also outperformed BO with a significantly larger budget in less than 35% of the total computation time.

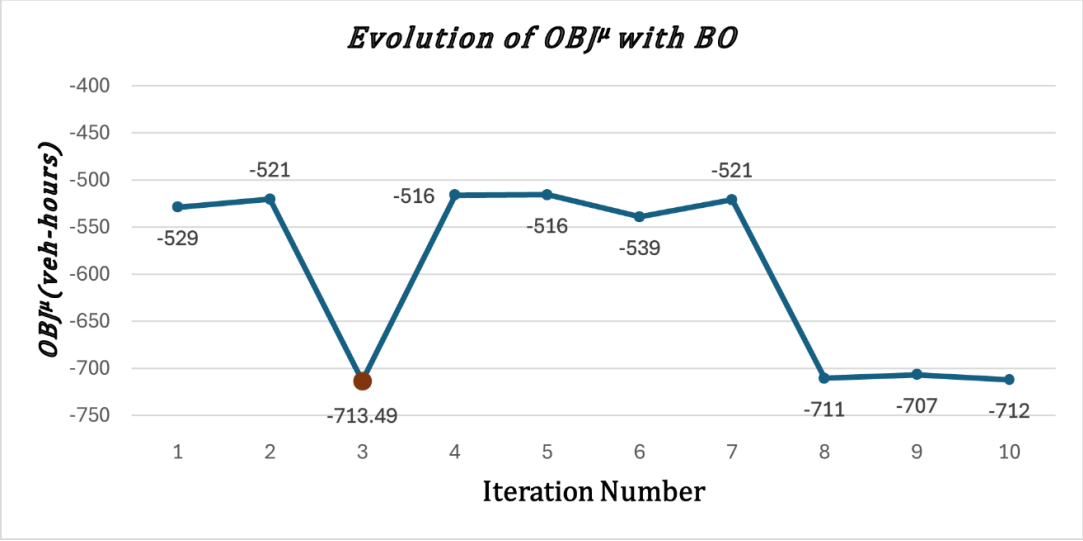



**Figure 14: Bayesian Optimization (BO) with 10 iterations applied to single-player problem with no external costs.**

*Variation of initial tolls*

To assess the sensitivity of the MBE scheme to its initial conditions, we conducted an additional experiment with initial tolls set to [0, 0, 0] hours. This variation was designed to test MBE's behavior when initialized close to the approximate global optimum. The results, illustrated in **Figure 15** and **Figure 16**, show that MBE scheme initially explores regions away from the starting point before returning to the optimal region by iteration 8. Notably, the MBE scheme revisits this region again in iterations 10 and 12 (not shown), demonstrates MBE scheme's robustness to initial conditions.

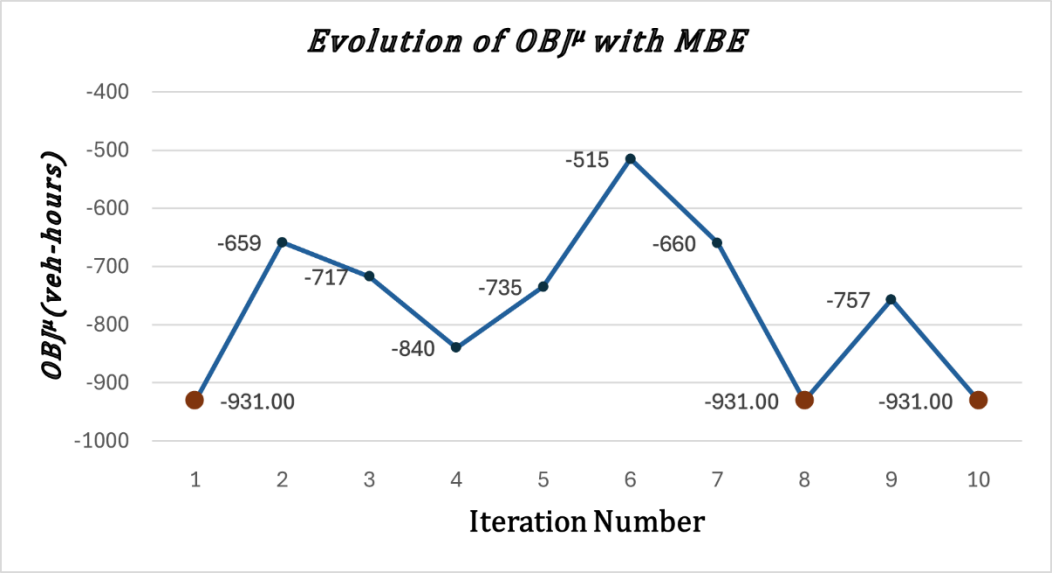



**Figure 15: Metamodel-based equilibration (MBE) scheme applied to single-player problem with no external costs with different initial tolls.**

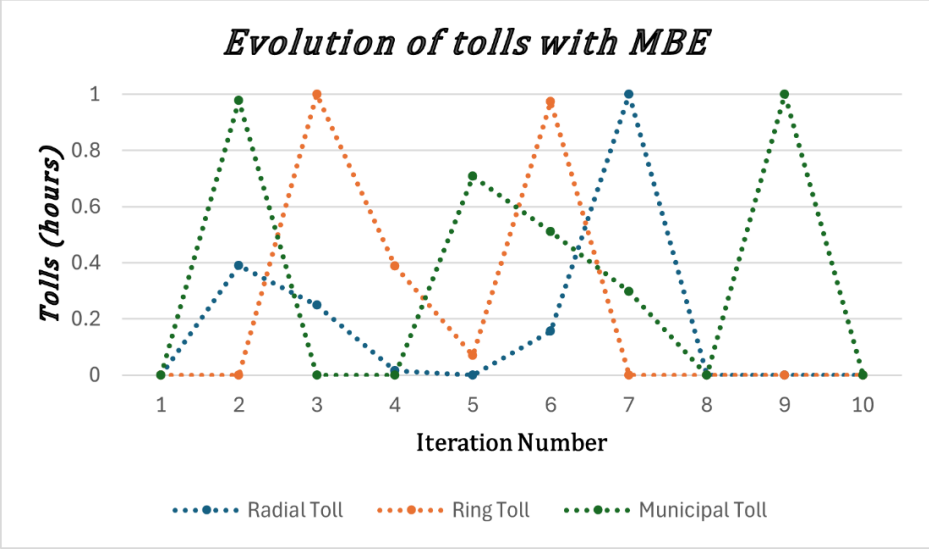



**Figure 16: Metamodel-based equilibration (MBE) scheme applied to single-player problem with no external costs with different initial tolls.**

*Standard external costs and high demand*

For this we go back to standard external costs and double the demand using **Equation (14)**:

$$\begin{aligned} new\ \boldsymbol{B_{RS}} &= \frac{\boldsymbol{B_{RS}}}{2} \\ new\ \boldsymbol{D_{RS}^{ref}} &= 2.\boldsymbol{D_{RS}^{ref}} \\ new\ \boldsymbol{D_{RS}^{max}} &= 2.\boldsymbol{D_{RS}^{max}} \end{aligned} \tag{14}$$

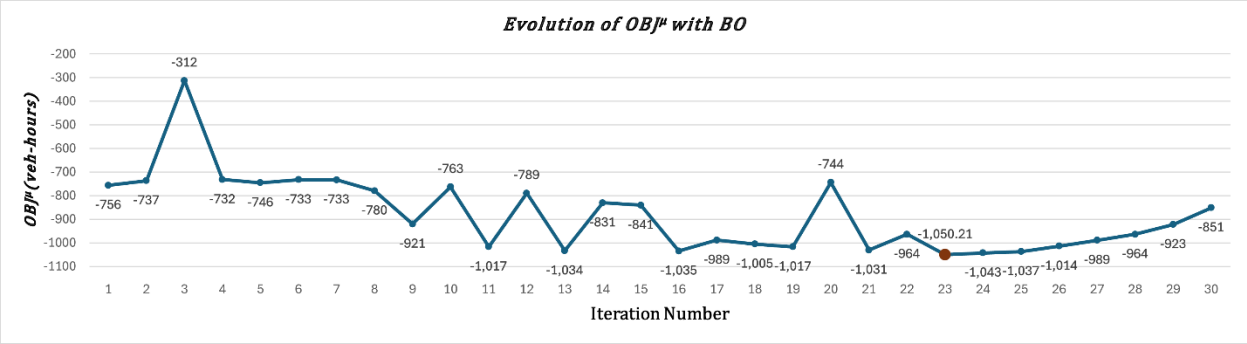



**Figure 17: Bayesian Optimization (BO) applied to single-player problem with standard external costs and high demand**.

BO with 30 iterations identified best $OBJ^{\mu}$ value of -1050.21 vehicle-hours at toll values of [0.01, 0.04, 0.76] hours (see **Figure 17**). The higher toll levels are consistent with the increased demand pressures and social external costs.

The evolution of $OBJ^{\mu}$ and tolls with MBE scheme are shown in **Figure 18** and **Figure 19** respectively, whereas the evolution of $OBJ^{\mu}$ for BO with 10 iterations is shown in **Figure 20**. The MBE scheme was executed for 7 iterations (5.09 hours) instead of 8 to enable a fairer comparison with the 10-iteration BO run (5.20 hours). The best MBE solution yielded an $OBJ^{\mu}$ value of -1077.86 vehicle-hours at tolls of [0.175, 0.00, 0.089] hours. Our MBE outperforms both the BO with 10 iterations (-921.81 vehicle-hours) and BO with 30 iterations (-1050.21 vehicle-hours).

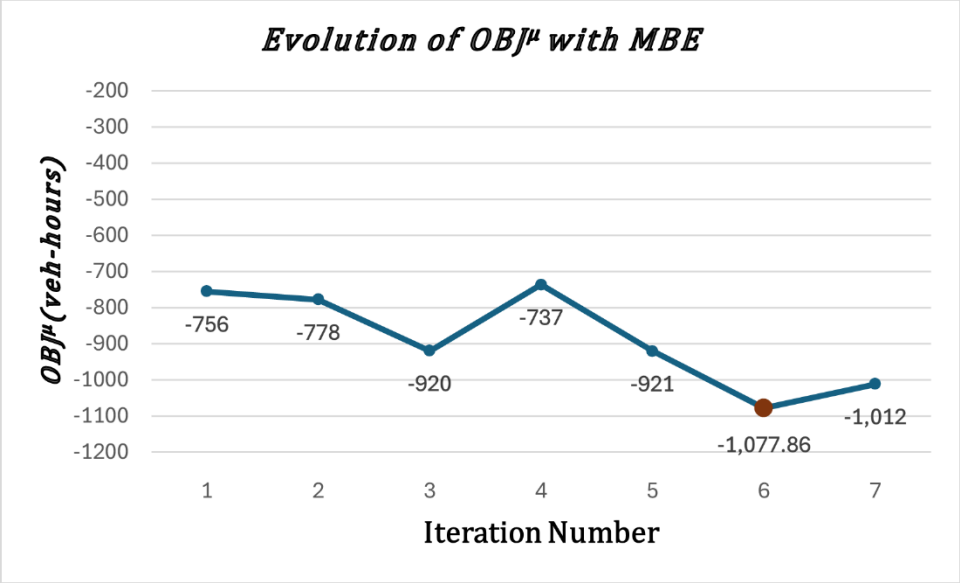



**Figure 18: Metamodel-based equilibration (MBE) scheme applied to single-player problem with standard external costs and high demand.**

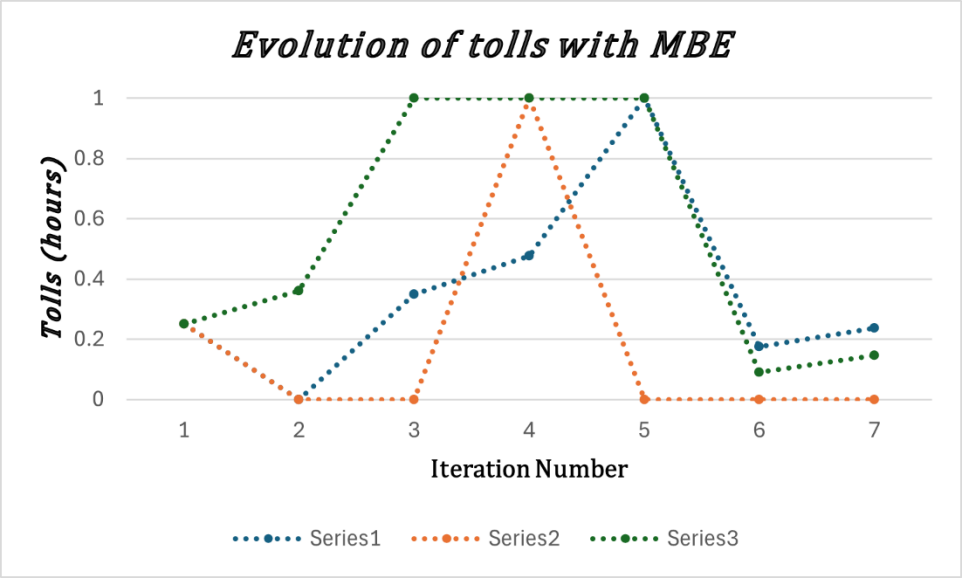



**Figure 19: Metamodel-based equilibration (MBE) scheme applied to single-player problem with standard external costs and high demand.**

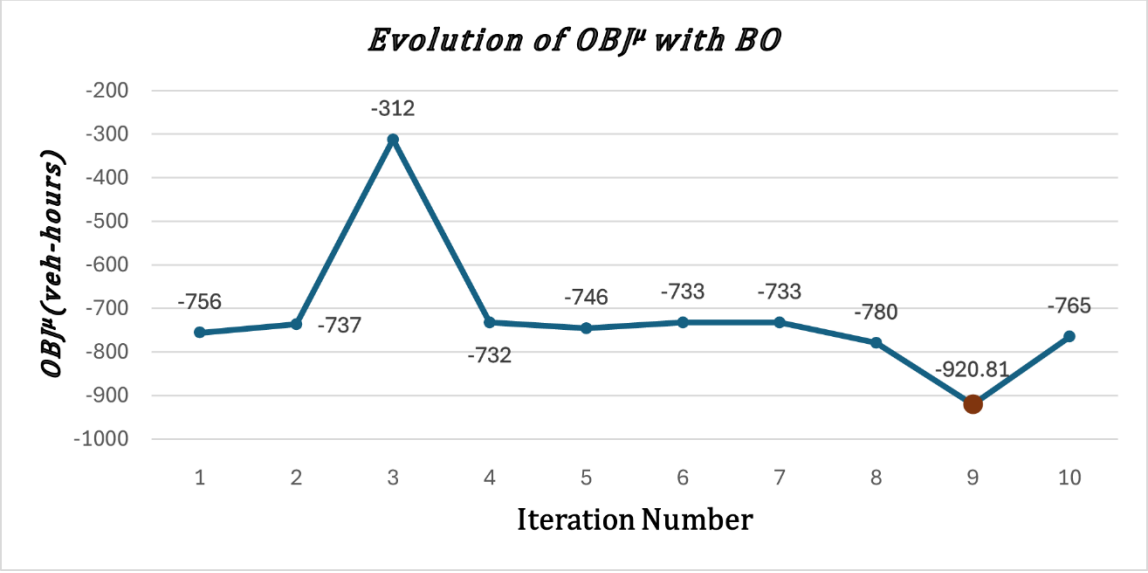



**Figure 20: Bayesian Optimization (BO) with 10 iterations applied to single-player problem with standard external costs and high demand**.

**Multi-player problem**

We consider two decision makers: (i) the city government (controlling $t_{rad} = t_{ring}$) and (ii) the surrounding municipalities (controlling $t_m$). Only the metamodel needs adaptation for multi-player problems; the underlying setup remains unchanged (except for equilibrium validation). This is the crucial advantage of MBE scheme that allows the solution of multiplayer problems in relatively manageable computational budgets compared to BO. BO-based benchmarking remains infeasible due to computational limits.

*Nash game*

We validate Nash equilibria (NE) by computing $\varepsilon_{\mu+}$ and $\varepsilon_{\mu-}$ of **Equation (13)** for both the city government and the surrounding municipalities, using a toll perturbation of $\phi = \pm 0.025$ hours from equilibrium values identified by the metamodel, and an $\varepsilon$ value of 1.6 vehicle-hours. **Figure 21** shows the evolution of $\varepsilon_{\mu+}$ and $\varepsilon_{\mu-}$ over the 10 iterations of the MBE scheme, where $\mu = 1$ refers to the city government and $\mu = 2$ refers to the surrounding municipalities. For better visualization, $\varepsilon_{\mu}$ values are clipped between -4 and +4. **Figure 21** shows four numerical ε-local NE, each with all four $\varepsilon_{\mu} <= 1.6$. The results for these four numerical ε-local NE are mentioned in **Table 1**. At these points, no unilateral perturbation of 0.025 hours by any decision maker would improve its objective function by more than 1.6 vehicle-hours. Notably, MBE scheme identified a valid numerical ε-local NE as early as in iteration 2. Comparing the four NE, NE2 (3rd iteration) stands out with three non-positive $\varepsilon_{\mu}$ and the remaining $\varepsilon_{1+}$ of 0.64 vehicle-hours.

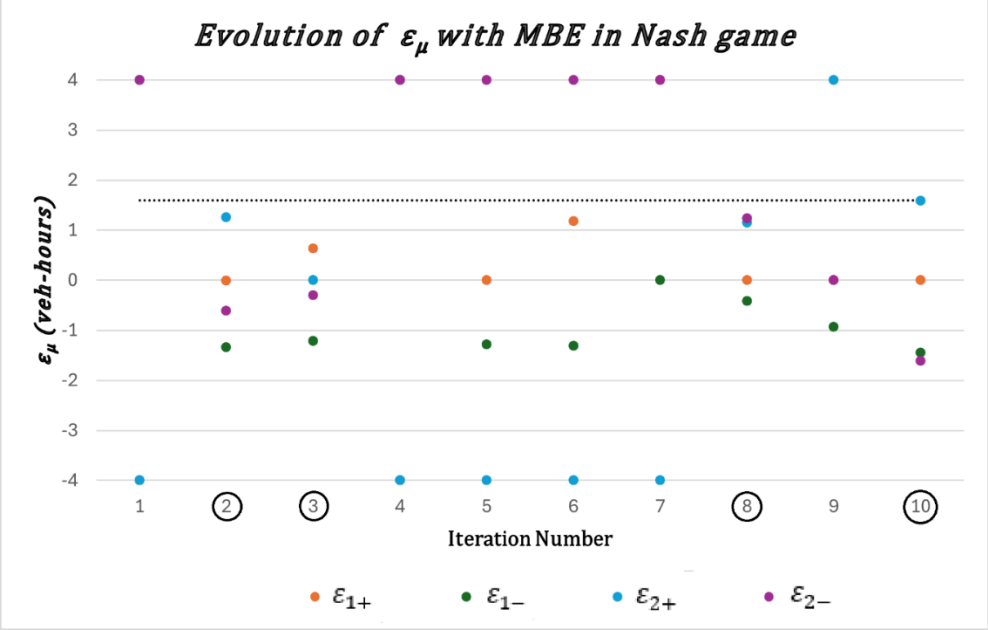



**Figure 21: Evolution of $\varepsilon_{\mu}$ with MBE scheme in Nash game (numerical ε-local Nash equilibria iterations are circled).**

**Table 1: Numerical ε-local Nash equilibria (NE) identified by MBE scheme.**

| Metric \ NE | NE1 | NE2 | NE3 | NE4 |
|---|---|---|---|---|
| **Iteration** | 2 | 3 | 8 | 10 |
| **$OBJ^1$ (veh. hr.)** | -53.4 | -60.6 | -54.2 | -24.0 |
| **$OBJ^2$ (veh. hr.)** | -688.2 | -688.2 | -687.6 | -717.1 |
| **$[t_{rad}, t_{ring}]$, $[t_m]$ (hr.)** | [1.00,1.00], [0.33] | [0.58,0.58], [1.00] | [1.00,1.00], [0.31] | [1.00,1.00], [0.83] |
| **$[\varepsilon_{1+}, \varepsilon_{1-}]$ (veh. hr.)** | [0.00, -1.33] | [0.64, -1.21] | [0.00, -0.41] | [0.00, -1.44] |
| **$[\varepsilon_{2+}, \varepsilon_{2-}]$ (veh. hr.)** | [1.25, -0.61] | [0, -0.29] | [1.15, 1.24] | [1.58, -1.62] |
| **Computation Time (hr.)** | 33.82 | | | |

The 10 MBE iterations required 33.82 hours due to sequential execution of four extra runs per iteration for computing $\varepsilon_{\mu+}$ and $\varepsilon_{\mu-}$. With infrastructure allowing parallelization, this could reduce to under 8 hours.

*Stackelberg game*

Validating Stackelberg equilibria (SE) is more complex than NE, as the valid response of the followers may itself change when the Stackelberg leader's toll is perturbed. Similar to the Nash game, we compute $\varepsilon_{2+}$ and $\varepsilon_{2-}$ for the follower decision maker (the surrounding municipalities) using a $\phi$ value of 0.025 hours. A follower's response is considered $\varepsilon$-valid if ε values remain below 1.6 vehicle-hours. **Figure 22** shows the evolution of $\varepsilon_{2+}$ and $\varepsilon_{2-}$ over the 10 iterations of the MBE scheme. Iterations 2 and 4 yield ε-valid follower responses; iteration 4 is strictly valid, with both ε values well below 0, indicating no incentive to change $t_m$ from 0.07 hours, when the leader (the city government) sets $t_{rad}$ and $t_{ring}$ of 1 hour. These two cases are reported as candidate SE in **Table 2**. SE2 (iteration 4) is selected as the most promising SE as it has more favorable value of Stackelberg leader's objective function i.e., $OBJ^1$

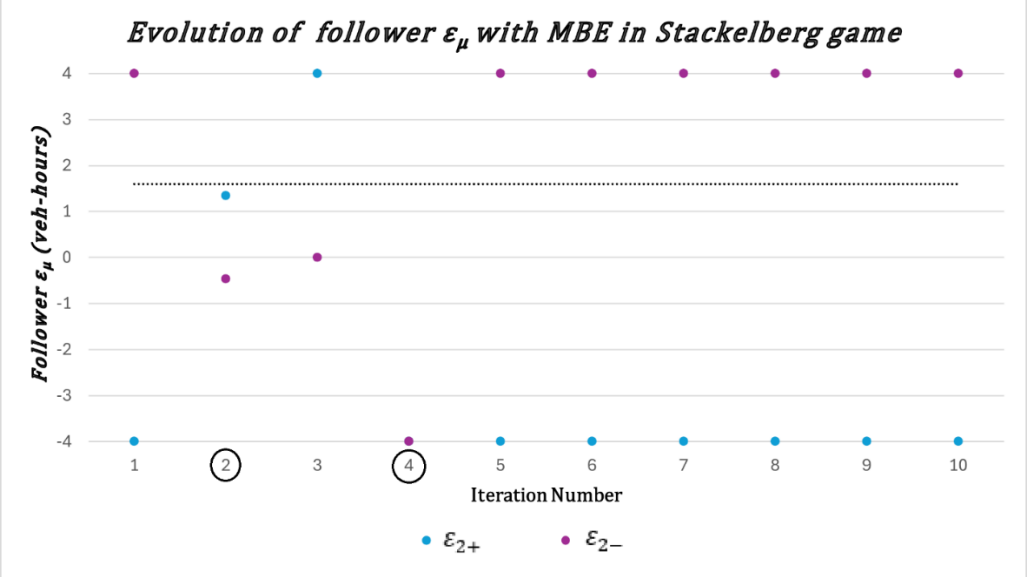



**Figure 22: Evolution of $\varepsilon_\mu$ with MBE scheme in Stackelberg game (iterations with a valid follower's response are circled).**

**Table 2: Candidate Stackelberg equilibria (SE) identified by MBE scheme.**

| Metric \ Candidate SE | SE1 | SE2 |
|---|---|---|
| **Iteration** | 2 | 4 |
| **$OBJ^1$ (veh. hr.)** | -53.4 | -63.06 |
| **$OBJ^2$ (veh. hr.)** | -688.1 | -806.74 |
| **$[t_{rad}, t_{ring}]$, $[t_m]$ (hr.)** | [1.00,1.00], [0.33] | [1.00,1.00], [0.07] |
| **$[\varepsilon_{2+}, \varepsilon_{2-}]$ (veh. hr.)** | [1.34, -0.47] | [-15.79, -38.37] |
| **Computation Time (hr.)** | 22.96 | |

It is noteworthy that in candidate SE2, both the Stackelberg leader and the follower achieve better objective function values compared to the candidate SE1 or any of the four numerical ε-local NE. This highlights that not all equilibria are equally desirable; some offer mutually superior outcomes for the involved decision makers. These findings underscore the importance of detailed multi-player analyses and motivate the development of methodological tools capable of addressing such problems, despite the inherent challenges.

## CONCLUSIONS

This paper introduces a novel metamodel-based equilibration (MBE) scheme for determining optimal or equilibrium tolls in multi-decision-maker pricing scenarios using full-scale traffic models with variable demand. The method employs a fast game-theoretical metamodel, built on the CHANAkYA framework, to shift toll equilibration from the computationally expensive traffic model to a simplified representation of the entire transportation system, enabling efficient multi-player analysis.

Unlike typical metamodel-based methods that approximate a single decision maker's objective, our metamodel captures the entire system and embeds strategic interactions among decision makers, allowing multi-player game-theoretical equilibrium to be computed through a single metamodel.

We apply the scheme to single- and two-player tolling scenarios on a real-world network in Aarschot, Belgium. In all cases, the MBE scheme outperforms benchmark Bayesian Optimization in both toll quality and computational efficiency. To our knowledge, this is the first demonstration of approximate Nash and Stackelberg equilibria computed via a directly interfaced elastic demand model and a full-scale STA. These results confirm that calibrated transport economics and mathematical programming models can yield robust tolling strategies in full-scale detailed traffic models.

## AUTHOR CONTRIBUTIONS

The authors confirm contribution to the paper follows: Conceptualization: Gaurav Malik, Chris M.J. Tampère; Everything else: Gaurav Malik. All authors reviewed the results and approved the final version of the manuscript

## DECLARATION OF CONFLICTING INTERESTS

The authors declared no potential conflicts of interest with respect to the research, authorship, and publication of this article.

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